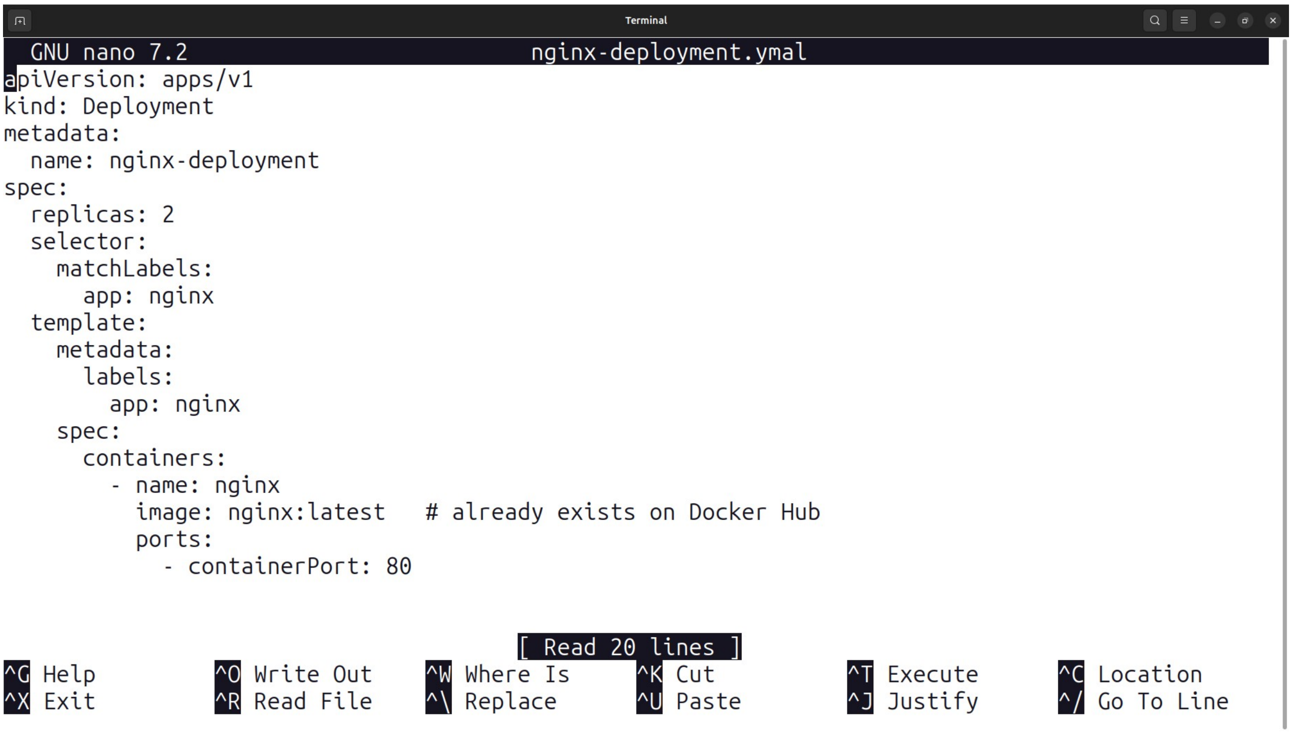
# Program7-Documentation

**Title:** Deploy a Web Application in Kubernetes using Minikube

This document outlines the steps to deploy a simple Nginx web server application onto a local Kubernetes cluster using Minikube.

### 1. Define the Deployment

The first step is to define the desired state of your application using a Kubernetes Deployment manifest file (typically a `.yaml` file).



```
Terminal
  GNU nano 7.2                          nginx-deployment.ymal
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest   # already exists on Docker Hub
          ports:
            - containerPort: 80


                                      [ Read 20 lines ]
^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute       ^C Location
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line
```

Key Components:

- kind: Deployment: Defines this object as a Deployment.
- replicas: 2: Ensures two identical instances (Pods) of the application will run.
- image: nginx:latest: Uses the official Nginx Docker image.
- containerPort: 80: The Nginx server inside the container listens on port 80.

### 2. Minikube start and status

Ensure your local Minikube cluster is running and configured correctly.

```
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ ls
nginx-deployment.ymal
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ nginx-deployment.ymal
nginx-deployment.ymal: command not found
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ nano nginx-deployment.ymal
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ minikube start
😄  minikube v1.37.0 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🏃  Updating the running docker "minikube" container ...
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass

❗  /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
    ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
```

### 3. Apply the deployment

Use kubectl to apply the configuration file and create the Deployment and its associated Pods in the Kubernetes cluster.

Command:

kubectl apply -f nginx-deployment.ymal

Verification:

deployment.apps/nginx-deployment created

### 4. Verify Deployment and Pod Status

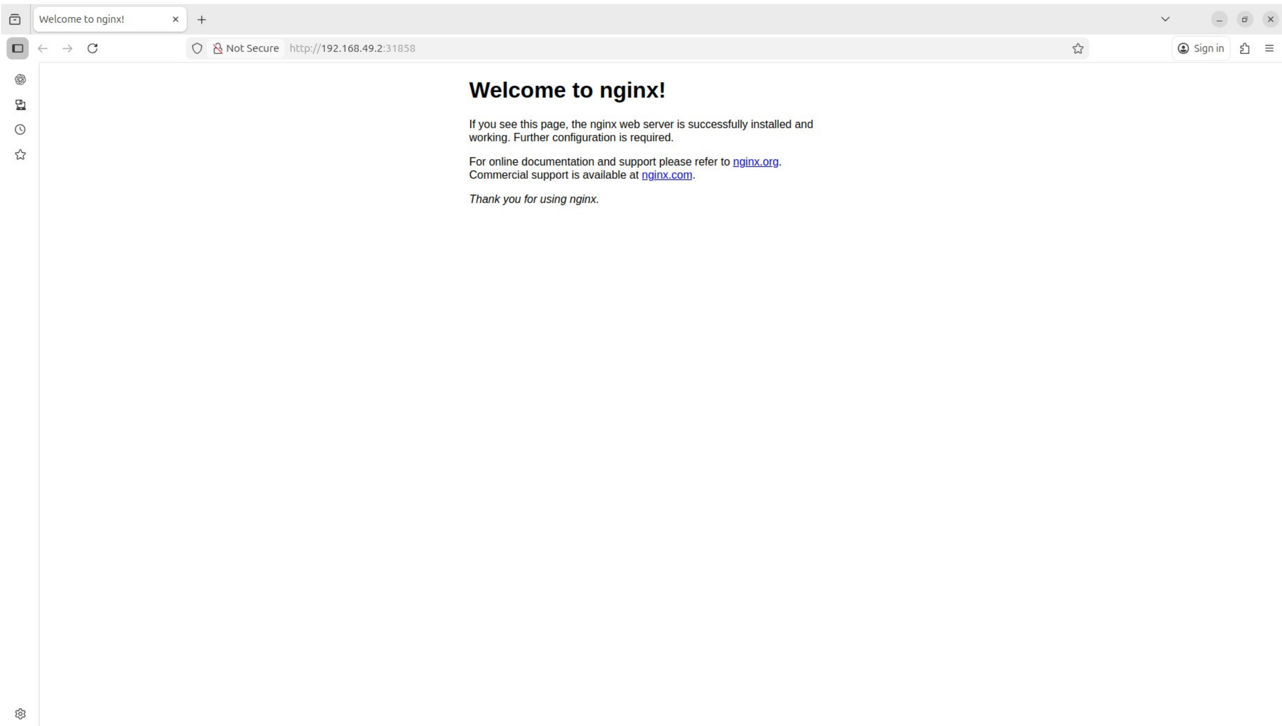
Check that the Deployment was successfully created and that the Pods are starting up.

```
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ kubectl apply -f nginx-deployment.ymal
deployment.apps/nginx-deployment created
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   0/2     2            0           19s
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ kubectl get pods
NAME                                READY   STATUS              RESTARTS      AGE
nginx-deployment-6f9664446b-p5t6h   0/1     ContainerCreating   0             23s
nginx-deployment-6f9664446b-w9p8m   0/1     ContainerCreating   0             23s
program-8                           1/1     Running             1 (78s ago)   43m
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ kubectl expose deployment nginx-deployment
--type=NodePort --port=80
service/nginx-deployment exposed
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ kubectl get services
NAME               TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes         ClusterIP   10.96.0.1       <none>        443/TCP        22h
nginx-deployment   NodePort    10.97.213.231   <none>        80:31858/TCP   18s
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ minikube service nginx-deployment
```

| NAMESPACE | NAME | TARGET PORT | URL |
|---|---|---|---|
| default | nginx-deployment | 80 | http://192.168.49.2:31858 |

```
🎉  Opening service default/nginx-deployment in default browser...
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program7$ ln: failed to create symbolic link '/home/1
```

## 5. Access the Web Application




Result (Terminal Output):

- The output provides the URL to access the application: http://192.168.49.2:31858
- Minikube attempts to open this URL in the default browser.

Final Verification (Browser):

- Navigating to the provided URL successfully displays the "Welcome to nginx!" page, confirming the web application is deployed and accessible.